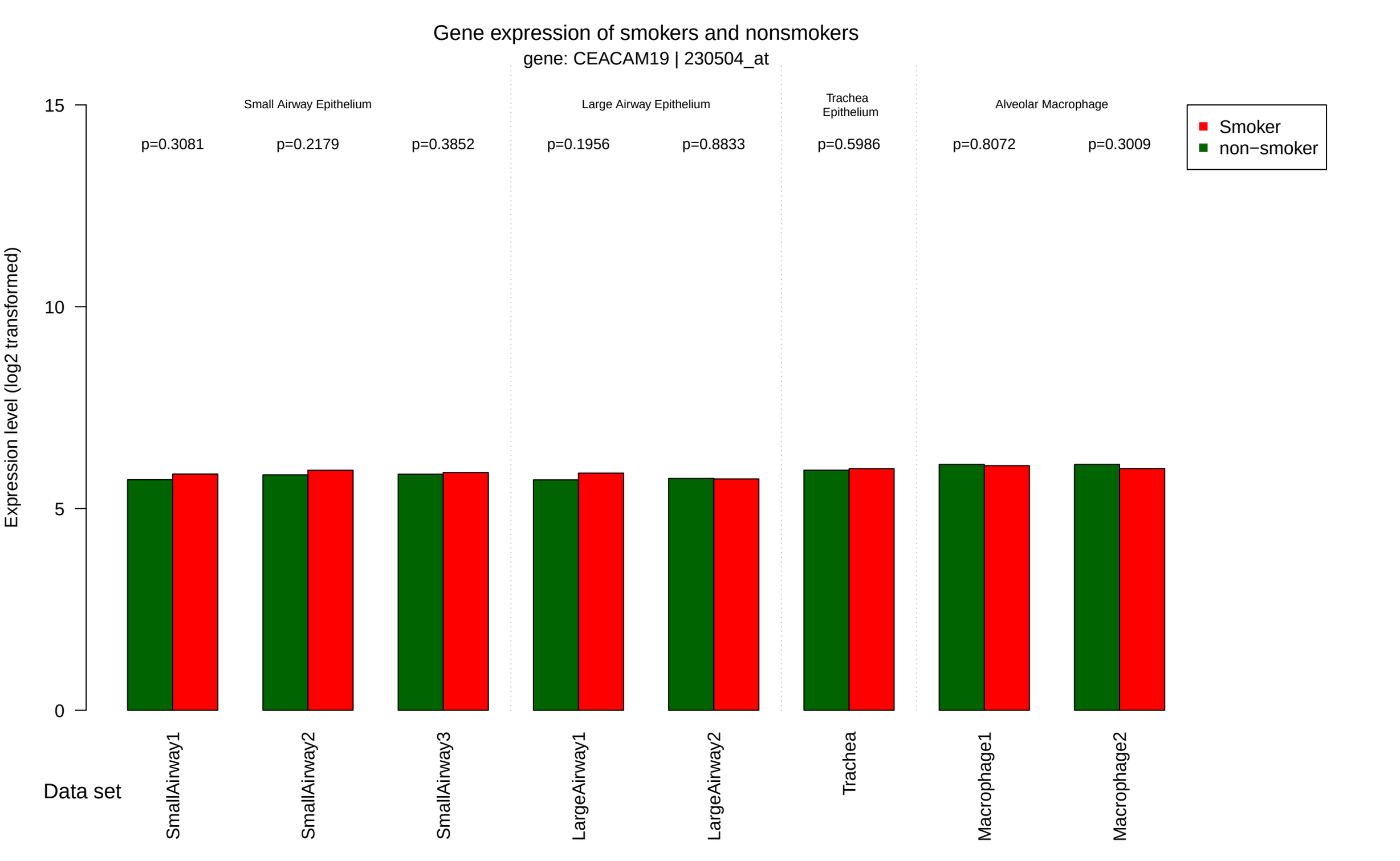
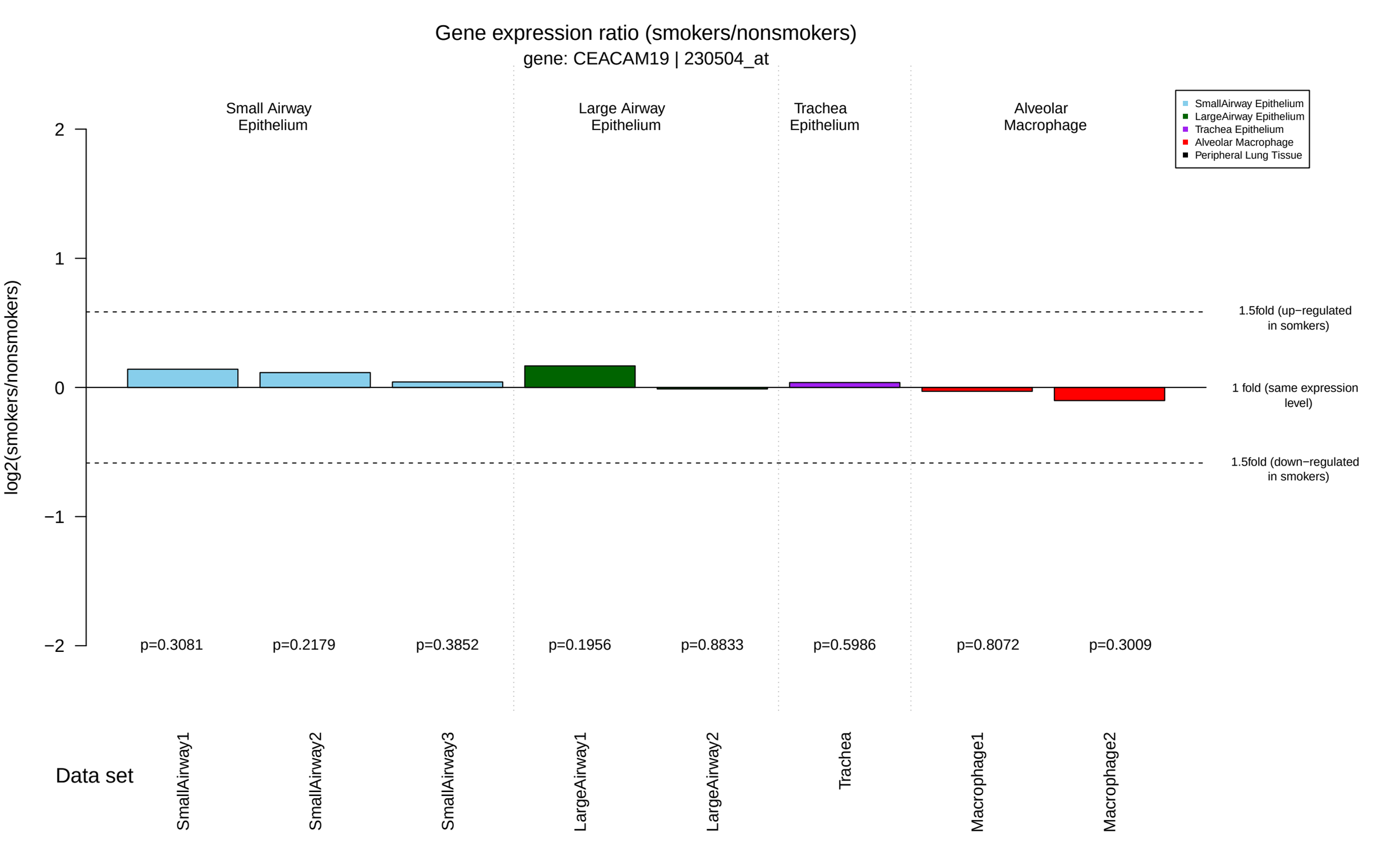
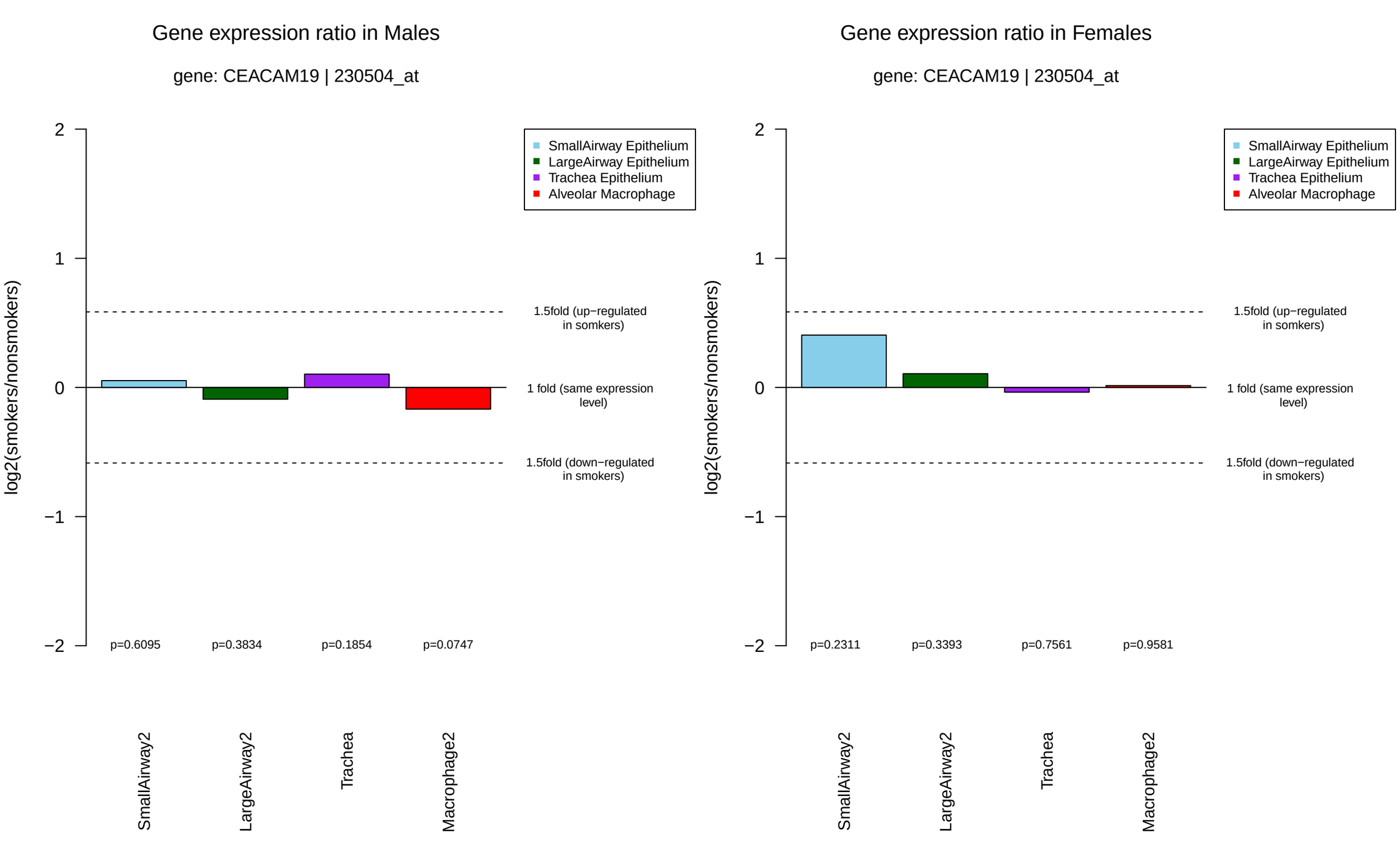
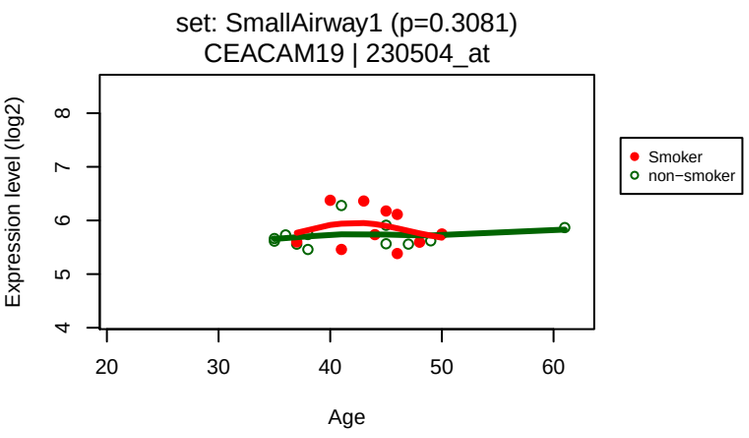
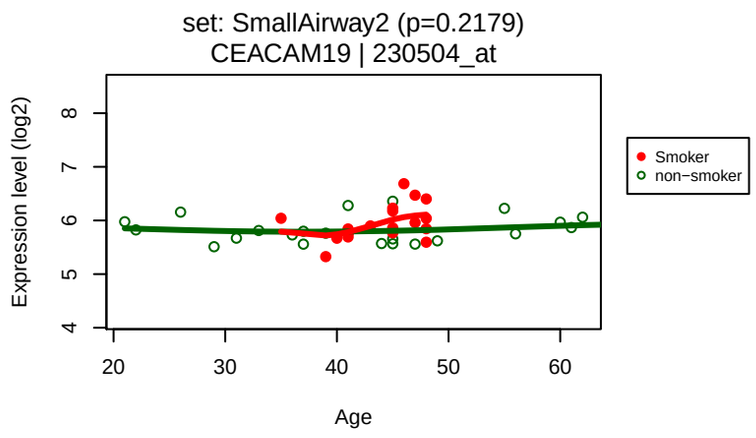
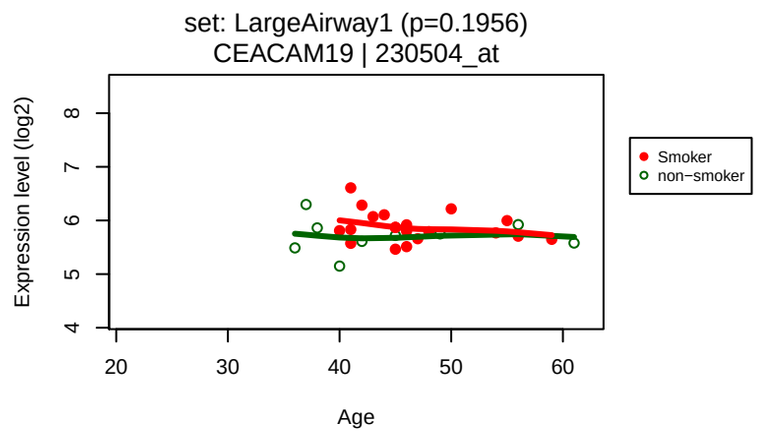

Gene expression of smokers and nonsmokers
gene: CEACAM19 | 230504_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3081
p=0.2179
p=0.3852
p=0.1956
p=0.8833
p=0.5986
p=0.8072
p=0.3009
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

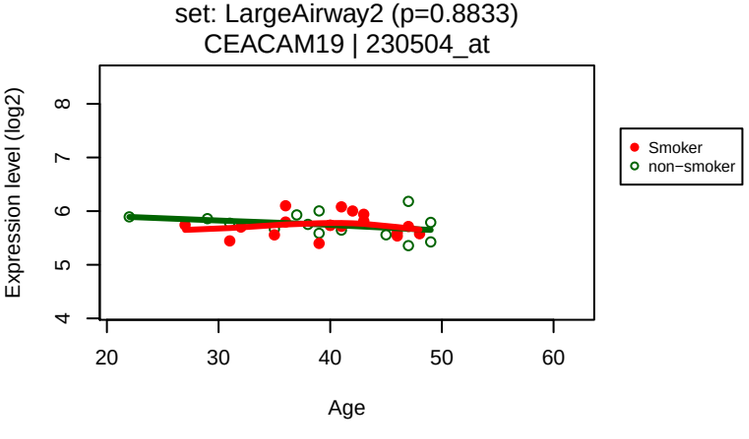

Gene expression ratio (smokers/nonsmokers)
gene: CEACAM19 | 230504_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3081
p=0.2179
p=0.3852
p=0.1956
p=0.8833
p=0.5986
p=0.8072
p=0.3009
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

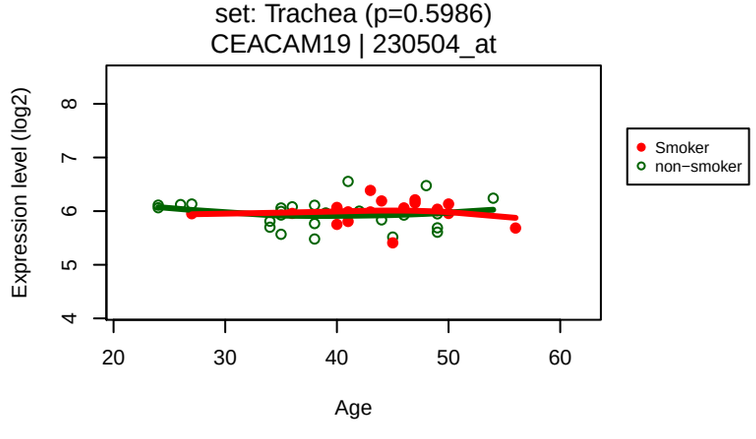

Gene expression ratio in Males
gene: CEACAM19 | 230504_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6095
p=0.3834
p=0.1854
p=0.0747
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CEACAM19 | 230504_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2311
p=0.3393
p=0.7561
p=0.9581
SmallAirway2
LargeAirway2
Trachea
Macrophage2

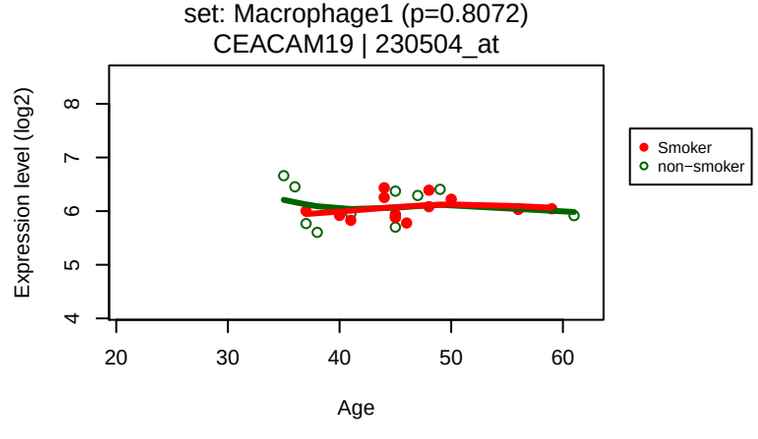

set: SmallAirway1 (p=0.3081)
CEACAM19 | 230504_at
Expression level (log2)
Age
Smoker
non−smoker

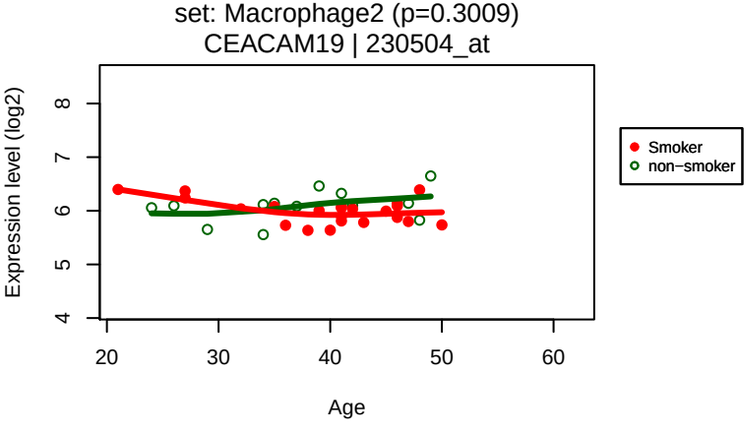

set: SmallAirway2 (p=0.2179)
CEACAM19 | 230504_at
Expression level (log2)
Age
Smoker
non−smoker

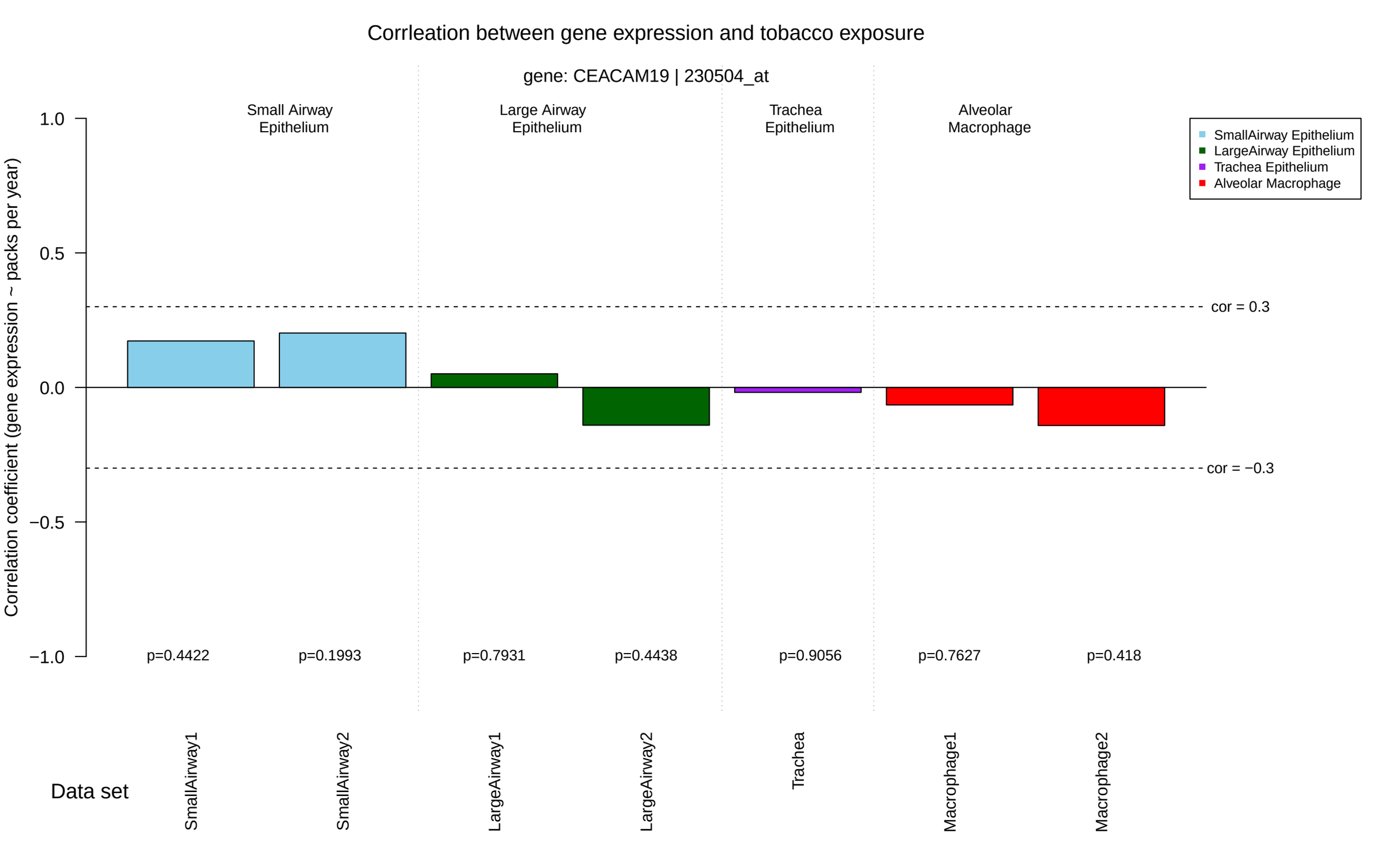

set: LargeAirway1 (p=0.1956)
CEACAM19 | 230504_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8833)
CEACAM19 | 230504_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5986)
CEACAM19 | 230504_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8072)
CEACAM19 | 230504_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3009)
CEACAM19 | 230504_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CEACAM19 | 230504_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4422
p=0.1993
p=0.7931
p=0.4438
p=0.9056
p=0.7627
p=0.418
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2